

Generation Interconnection System Impact Study Report

For Queue Project Q22-2115

Seminole 138 kV

May 29, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2115. The Interconnection Customer proposes a 637.6 MW solar photovoltaic facility interconnecting at the Seminole 138 kV substation in the Delaware area. The request was received on June 15, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 728 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$50,579,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Seminole - Milltown 138 kV line is overloaded to 300.1% of its 195 MVA emergency rating for the loss of the Seminole - Shred 138 kV line. The project contributes 100.0% DFAX (637.6 MW) to this overload.

Seminole - Shred 138 kV: 291.8% loading (200 MVA) following the outage of the Milltown - Seminole 138 kV; DFAX = 100.0%.

The Milltown - Kiamensi 138 kV is overloaded to 291.1% of its 198 MVA emergency rating for the loss of the Shred - Seminole 138 kV Line. The project contributes 100.0% DFAX (637.6 MW) to this overload.

Otay Tap - Shred 138 kV: 251.1% loading (220 MVA) following the outage of the Milltown - Seminole 138 kV line; DFAX = 100.0%.

Beacon - Kiamensi Substation 138 kV: 223.3% loading (232 MVA) following the outage of the Seminole - Shred 138 kV; DFAX = 100.0%.

Denham Springs Substation - Star Peak 138 kV: 218.6% loading (242 MVA) following the outage of the Milltown - Seminole 138 kV; DFAX = 70.5%.

For the loss of Milltown - Seminole 138 kV Line, the Otay Tap - Denham Springs 138 kV line loads to 209.2% of its 249 MVA rating. Project DFAX: 100.0%.

Beacon - Broadview 138 kV L: 200.2% loading (243 MVA) following the outage of the Seminole - Shred 138 kV; DFAX = 100.0%.

The Harmony - Broadview 138 kV is overloaded to 150.0% of its 300 MVA emergency rating for the loss of the Alessandro - Carver Falls 138 kV. The project contributes 29.7% DFAX (189.2 MW) to this overload.

The Dell Creek - Glasgow 138 kV is overloaded to 144.0% of its 218 MVA emergency rating for the loss of the Seminole - Milltown 138 kV Line. The project contributes 13.3% DFAX (84.9 MW) to this overload.

Glasgow - Blooming Prairie 138 kV: 137.3% loading (255 MVA) following the outage of the Seminole - Milltown Substation 138 kV; DFAX = 13.3%.

Blooming Prairie - MillCreek 138 kV: 135.7% loading (264 MVA) following the outage of the Milltown - Seminole 138 kV L; DFAX = 13.3%.

Carver Falls - Alessandro 138 kV: 133.2% loading (236 MVA) following the outage of the Harmony - Broadview 138 kV L; DFAX = 17.6%.

The Maurice - Alessandro 138 kV is overloaded to 128.0% of its 242 MVA emergency rating for the loss of the Broadview - Harmony 138 kV. The project contributes 17.7% DFAX (112.6 MW) to this overload.

New Castle - Carver Falls 138 kV Line: 110.7% loading (249 MVA) following the outage of the Harmony - Broadview 138 kV Line; DFAX = 17.6%.

The Mill Cteek - Conoco S 138 kV is overloaded to 108.5% of its 182 MVA emergency rating for the loss of the Denham Springs - Star Peak 138 kV. The project contributes 5.7% DFAX (36.2 MW) to this overload.

Perryville - Conoco S 138 kV: 108.2% loading (199 MVA) following the outage of the Star Peak - Denham Springs 138 kV Line; DFAX = 5.7%.

The Basin Rd - West 138 kV L is overloaded to 103.6% of its 186 MVA emergency rating for the loss of the Harmony - Broadview 138 kV. The project contributes 7.7% DFAX (49.2 MW) to this overload.

Springville - Waste Management Northern Oaks Lfgte 138 kV L: 102.1% loading (241 MVA) following the outage of the Denham Springs - Star Peak 138 kV; DFAX = 5.7%.

For the loss of Broadview Substation - Harmony 138 kV, the Maurice - Sunset Lake 138 kV loads to 100.3% of its 248 MVA rating. Project DFAX: 22.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Seminole - Milltown 138 kV line	Rebuild line with higher-capacity conductor	\$1,731,000	100% (\$1,731,000)
Seminole - Shred 138 kV L	Rebuild line with higher-capacity conductor	\$1,360,000	100% (\$1,360,000)
Milltown - Kiamensi 138 kV	Rebuild line with higher-capacity conductor	\$2,787,000	100% (\$2,787,000)
Oty Tap - Shred 138 kV	Rebuild line with higher-capacity conductor	\$2,477,000	100% (\$2,477,000)
Beacon - Kiamensi 138 kV	Rebuild line with higher-capacity conductor	\$5,039,000	100% (\$5,039,000)
Denham Springs Substation - Star Peak 138 kV	Rebuild line with higher-capacity conductor	\$1,465,000	100% (\$1,465,000)
Otay Tap - Denham Springs Substation 138 kV	Rebuild line with higher-capacity conductor	\$5,609,000	100% (\$5,609,000)
Beacon - Broadview 138 kV L	Rebuild line with higher-capacity conductor	\$1,939,000	100% (\$1,939,000)
Harmony - Broadview Substation 138 kV line	Rebuild line with higher-capacity conductor	\$2,045,000	71% (\$1,455,000)
Dell Crk - Glasgow 138 kV	Rebuild line with higher-capacity conductor	\$3,734,000	56% (\$2,087,000)
Blooming Prairie - Glasgow 138 kV	Rebuild line with higher-capacity conductor	\$14,547,000	56% (\$8,130,000)
Blooming Prairie Substation - Mill Cteek 138 kV Line	Rebuild line with higher-capacity conductor	\$4,357,000	56% (\$2,435,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Alessandro Substation - Carver Falls 138 kV Line	Rebuild line with higher-capacity conductor	\$5,586,000	57% (\$3,207,000)
Alessandro - Maurice 138 kV	Rebuild line with higher-capacity conductor	\$3,253,000	57% (\$1,868,000)
New Castle - Carver Falls 138 kV Line	Rebuild line with higher-capacity conductor	\$2,823,000	57% (\$1,621,000)
Conoco South - Mill Cteek 138 kV Line	Reconductor line	\$3,822,000	100% (\$3,822,000)
Conoco S - Perryville 138 kV	Reconductor line	\$808,000	100% (\$808,000)
Basin Rd - West 138 kV L	Reconductor line	\$933,000	50% (\$466,000)
Springville - Waste Manaement Northern Oaks Lfgte 138 kV	Reconductor line	\$1,643,000	100% (\$1,643,000)
Maurice - Sunset Lake 138 kV	Reconductor line	\$865,000	73% (\$630,000)
	Total allocated to project		\$50,579,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.